

Data Analyst Capstone Project

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03 July 2026



OUTLINE



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- Discussion
 - Findings & Implications
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EXECUTIVE SUMMARY



trends in programming languages and databases

Current Technology Used

- Top 10 Languages worked With
- Top 10 Data Bases Worked With
- Platform Worked With
- Top 10 Web-frame Worked With

• Future Technology Trend

- Top 10 languages desire Next Year
- Top 10 Databases Desired Next Year
- Platform worked with Desired Next Year
- TOP 10 Web frame desired next year
-

- DEMOGRAPHICS

- .

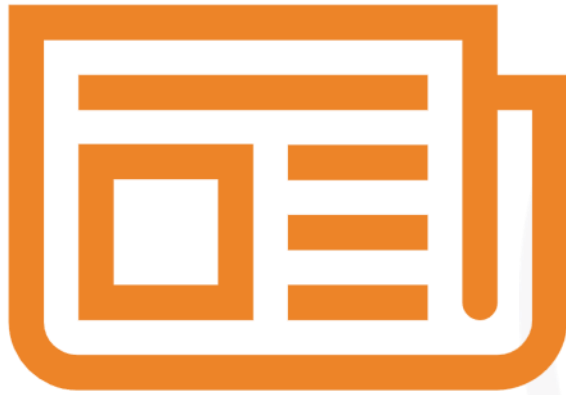
INTRODUCTION



Purpose of the report, Target Audience & Value of the analysis

- The point of the report is to make inferences about trends in programming languages and changes in the interest of developers over time.
- The target audience are people that make decisions on which databases and languages that developers are using and would like to use
- The value of the analysis is to give insights on decision making on what languages are key for developers, their hobbies, compensation and changes in the trends of languages being used

METHODOLOGY



Data sources, Data collection & Data wrangling steps

- -The data sources used for the project were obtained from
 - Job postings
 - Training portals
 - Developer surveys, such as the latest Stack Overflow Developer Survey

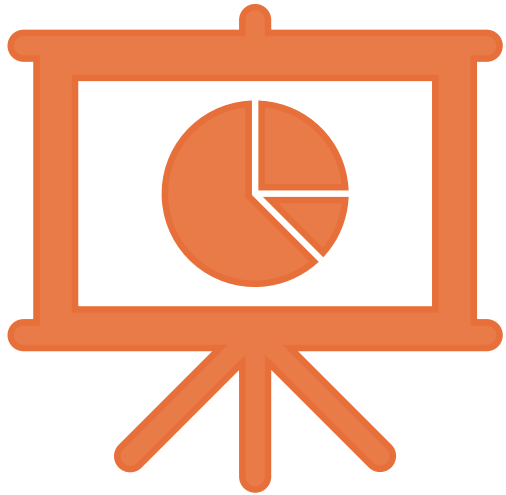
- **Data collection**

- -The data was collected through the use of web scrapping and making use of apis

Data wrangling

- The data wrangling techniques that were used was to ensure there are null columns .
- -The numeric columns that had null values were replaced by the mean of the data, for categorical columns we replaced null values by the mode of the column

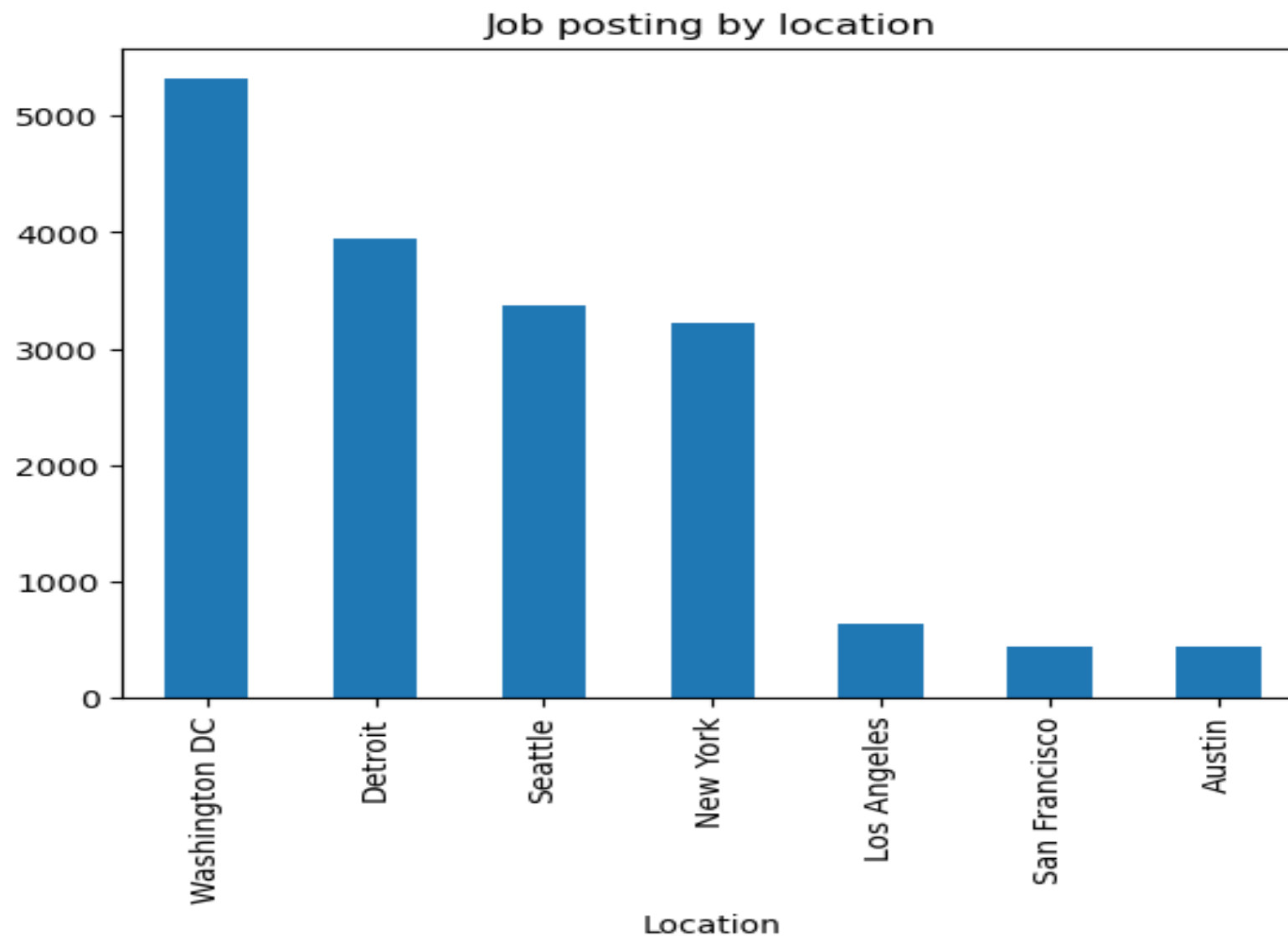
Results



<Please present your results in the following slides.>

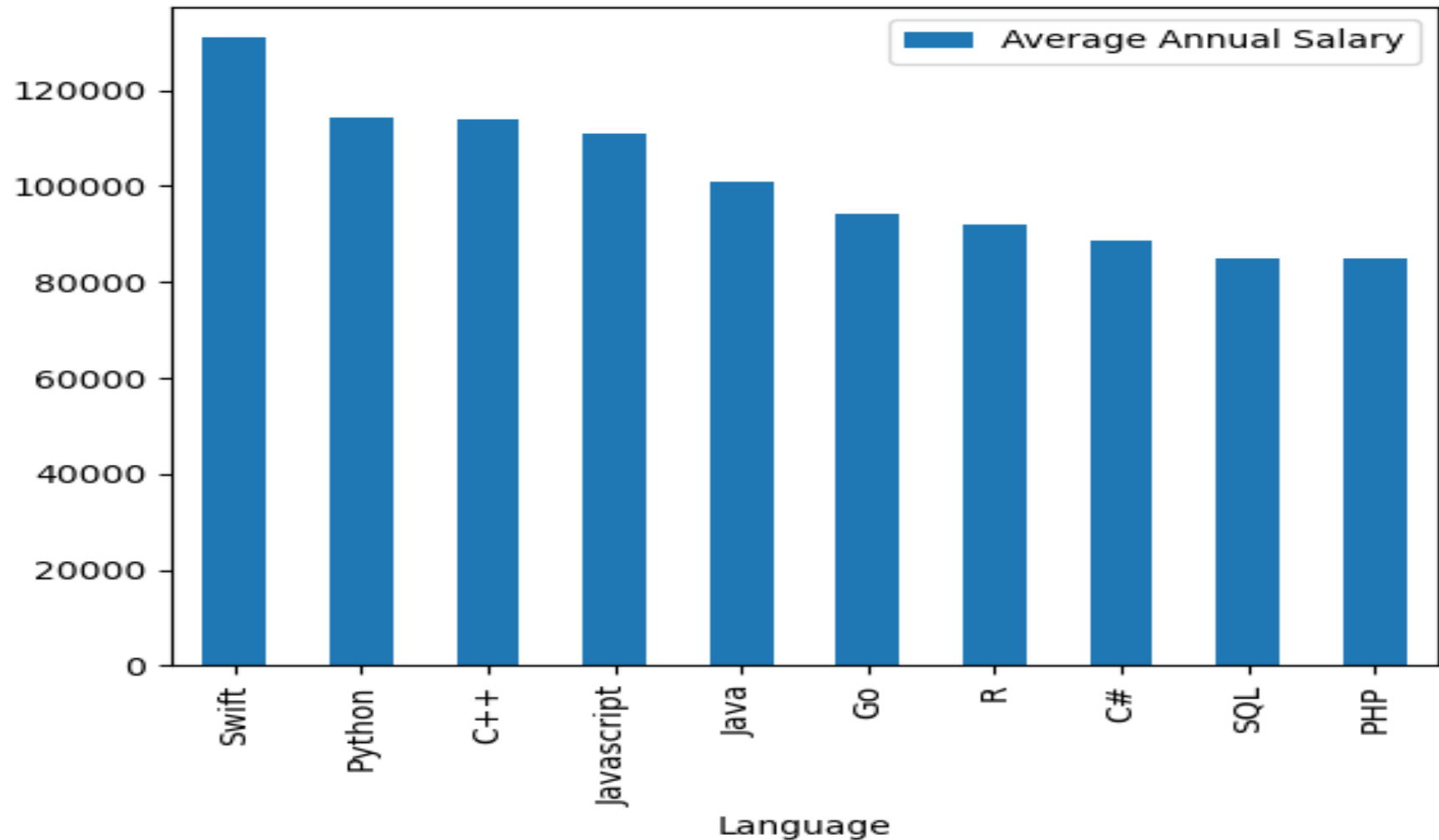


JOB POSTINGS



POPULAR LANGUAGES

I



PROGRAMMING LANGUAGE TRENDS

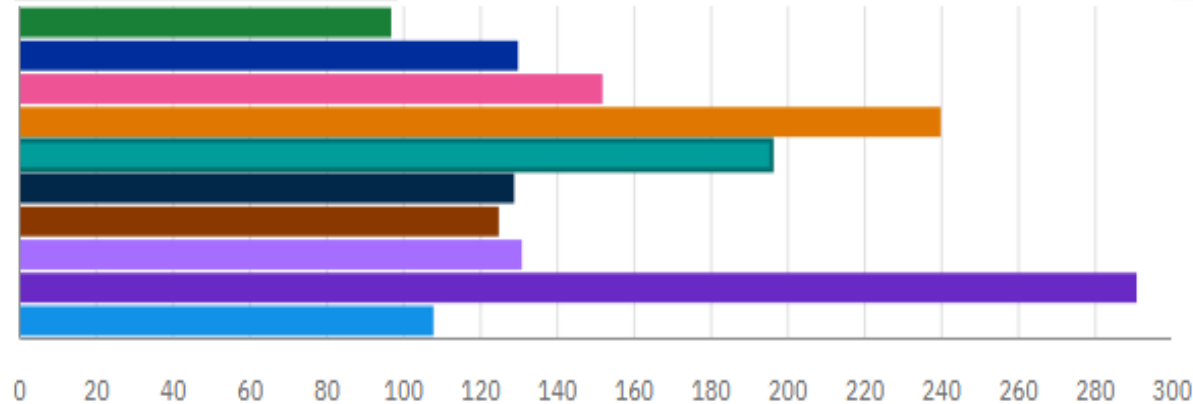
Summarize key trends shown in the charts.

Current Year

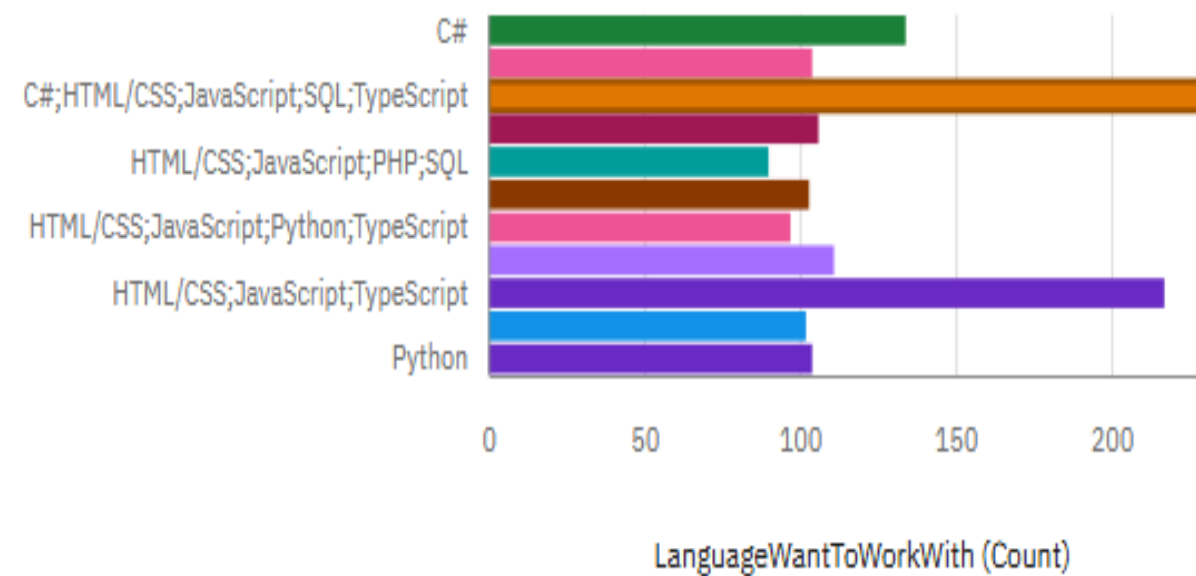
LanguageHaveWorkedWith by LanguageHaveWorkedWith colored by LanguageHaveWorkedWith

LanguageHaveWorkedWith

- C#
- C#;HTML/CSS;JavaScript;PowerS...
- C#;HTML/CSS;JavaScript;SQL
- C#;HTML/CSS;JavaScript;SQL;Ty...
- HTML/CSS;JavaScript;PHP;SQL
- HTML/CSS;JavaScript;PHP;SQL;T...



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript ,Sql and html are the top three languages
- The salaries of programming languages keep increasing year by year
- Python is going to be more popular next year

Implications

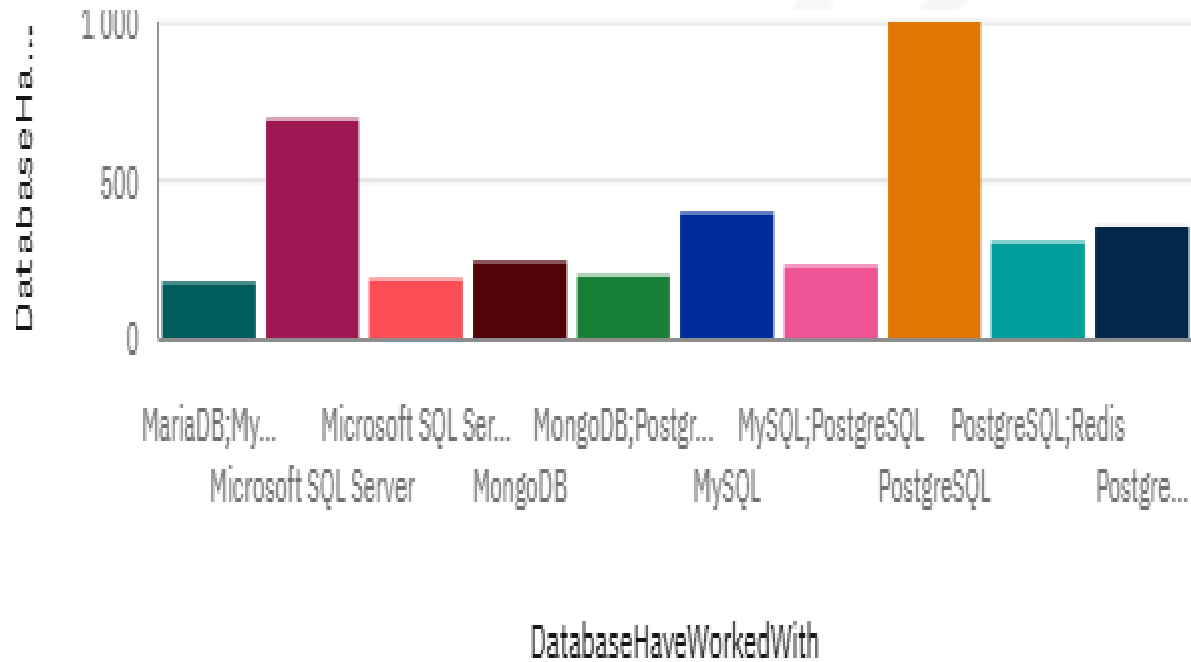
- Companies need to invest more in languages like python and Javascript and training to prepare their workers for in demand needed skills.
 - More people will move into programming and developing due to the increase in the salaries of the jobs.
- .There is need to look at extensive training in programming languages like python in businesses.



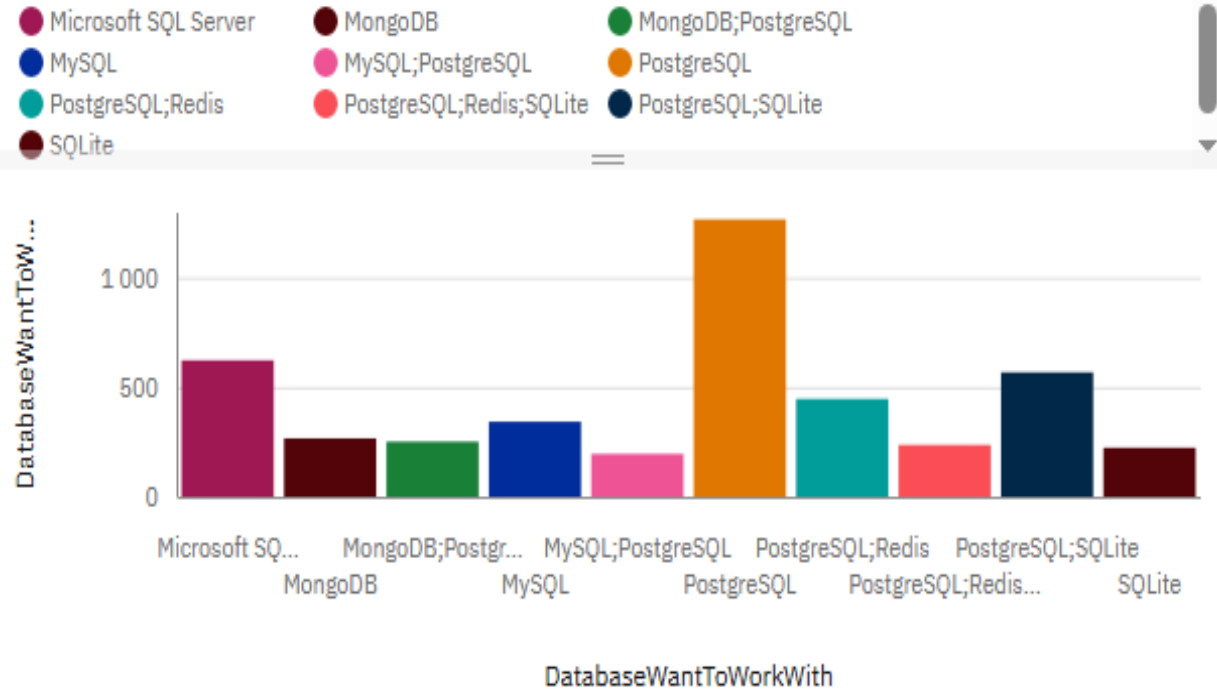
DATABASE TRENDS

Summarize key trends shown in the charts.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- The most popular databases are PostgreSQL and my sql
- The least popular databases are Mongo DB and Redis , but there are going to be more popular in the next year by comparing the two charts

Implications

- Postgre SQL is going to be a very important language in the coming years
- People still have preference for open- source databases compared to cloud databases in most companies



DASHBOARD



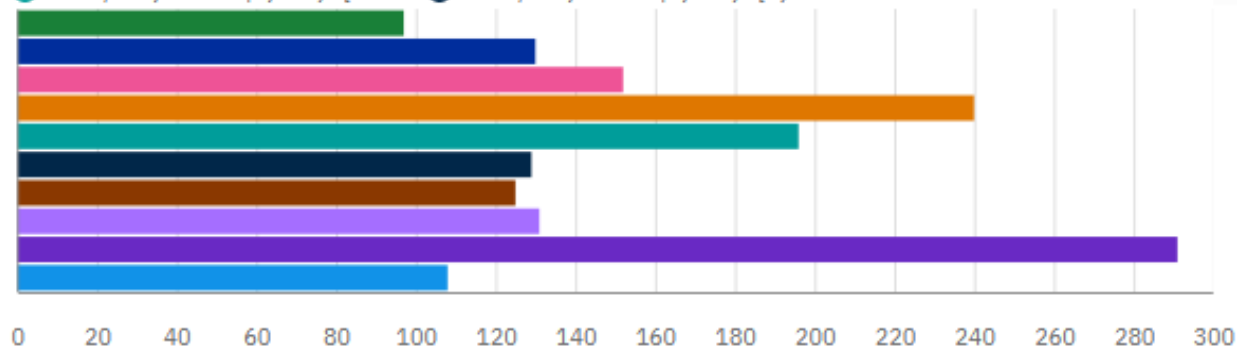
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LanguageHaveWorkedWith

LanguageHaveWorkedWith

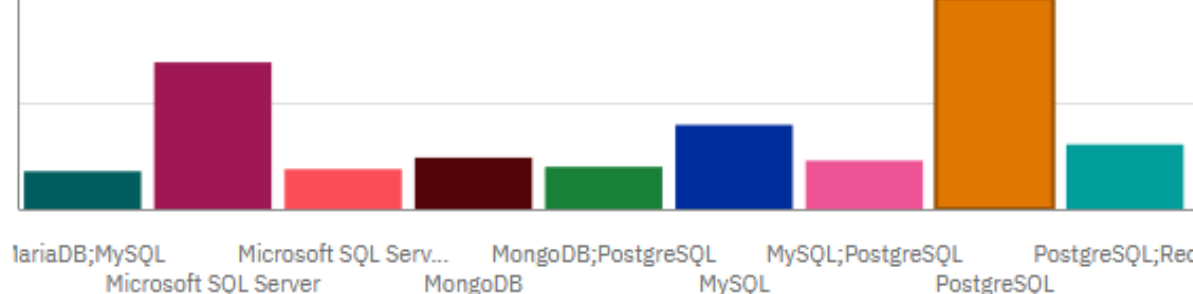
- C#
- C#;HTML/CSS;JavaScript;PowerS...
- C#;HTML/CSS;JavaScript;SQL
- C#;HTML/CSS;JavaScript;SQL;Ty...
- HTML/CSS;JavaScript;PHP;SQL
- HTML/CSS;JavaScript;PHP;SQL;T...



DatabaseHaveWorkedWith

DatabaseHaveWorkedWith

- MariaDB;MySQL
- Microsoft SQL Server
- MongoDB
- MuSQL - PostreSQL
- Microsoft SQL Server;PostgreSQL
- MongoDB;PostgreSQL
- MySQL
- MySQL;PostgreSQL
- PostgreSQL
- PostgreSQL;Red



PlatformHaveWorkedWith colored by PlatformHaveWorkedWith sized by PlatformHaveWorkedWith

PlatformHaveWor...



PlatformHaveWorkedWith

- Amazon Web Services (AWS)
- Amazon Web Services (AWS);Digi...
- Amazon Web Services (AWS);Goo...
- Amazon Web Services (AWS);Micr...
- Digital Ocean
- Google Cloud



WebframeHaveWorkedWith hierarchy colored by WebframeHaveWorkedWith and sized by WebframeHaveWorkedWith

WebframeHaveW...



WebframeHaveWorkedWith

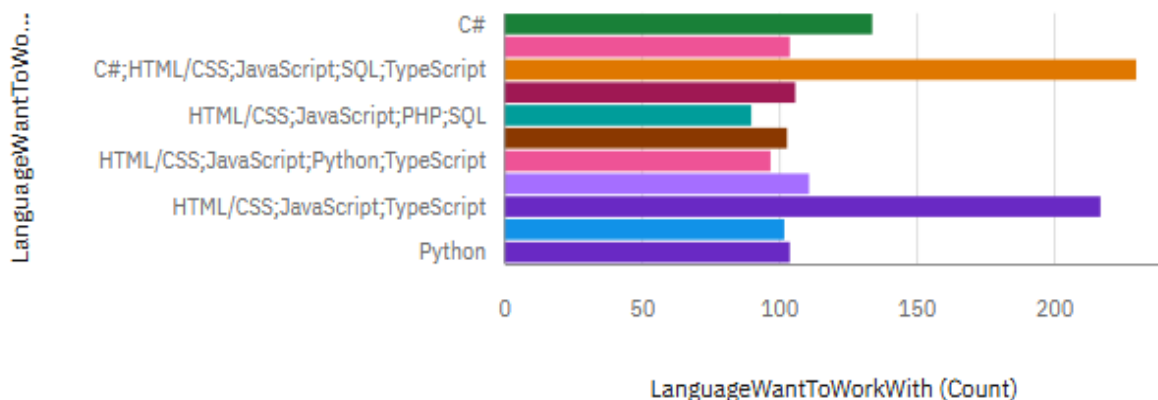
- ASP.NET CORE
- ASP.NET;ASP.NET CORE
- Express;Next.js;Node.js;React



LanguageWantoWorkWith Bar chart Top 10

LanguageWantoWorkWith

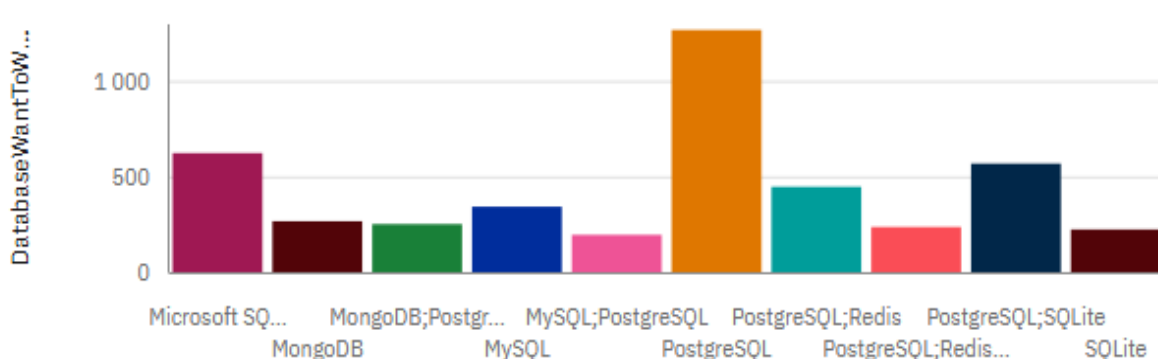
- C#
- C#;HTML/CSS;JavaScript;SQL;TypeScript
- HTML/CSS;JavaScript;PHP;SQL
- HTML/CSS;JavaScript;Python;TypeScript
- C#;HTML/CSS;JavaScript;SQL
- C#;SQL
- HTML/CSS;JavaScript;Python;SQL
- HTML/CSS;JavaScript;SQL;TypeScript
- Python



DatabaseWantoWorkWith Column Chart Top 10

DatabaseWantoWorkWith

- Microsoft SQL Server
- MySQL
- PostgreSQL;Redis
- SQLite
- MongoDB
- MySQL;PostgreSQL
- PostgreSQL;Redis;SQLite
- MongoDB;PostgreSQL
- PostgreSQL
- PostgreSQL;SQLite



PlatformWantoWorkWith Top 10

PlatformWantoW...

PlatformWantoW...



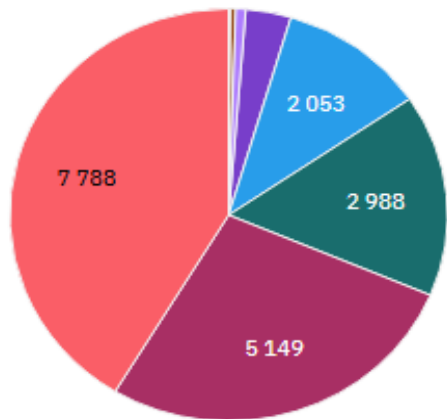
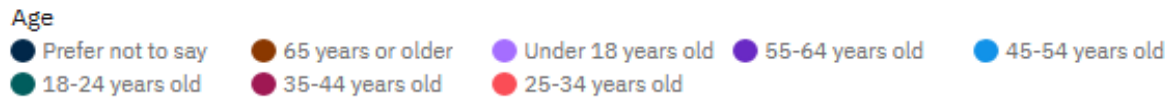
WebframeWantoWorkWith hierarchy top 10

WebframeWantoTo...



DASHBOARD TAB 3: Demographics

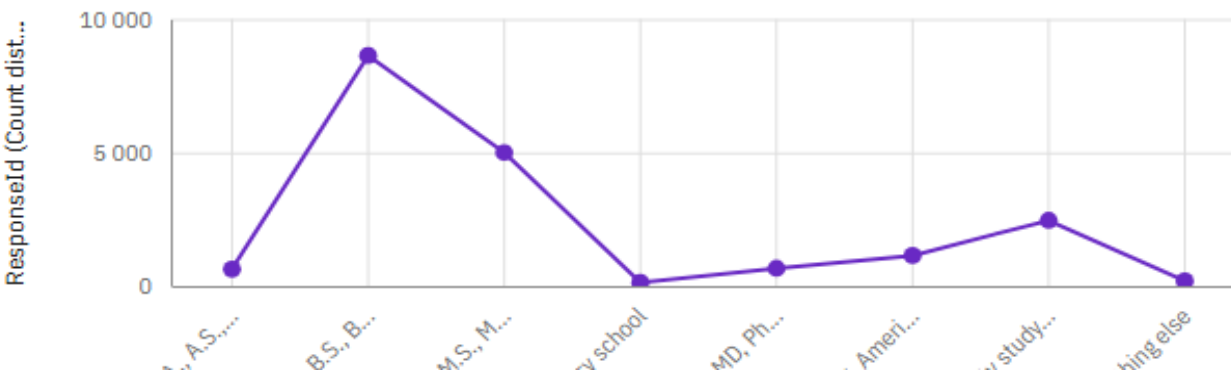
Respondent distribution by Age



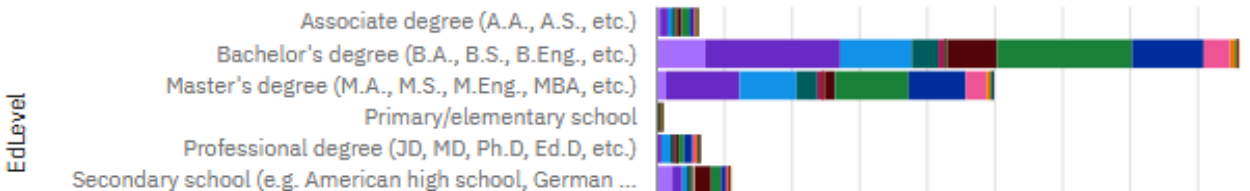
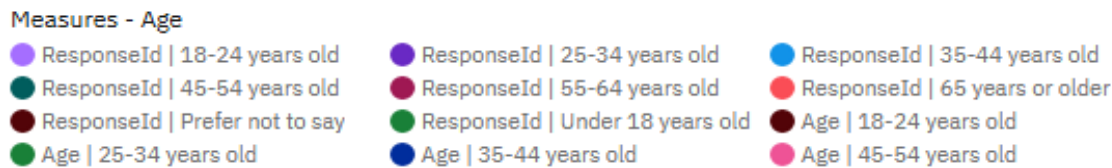
Country for ResponseId and Country regions



Response distribution by age



Respondent Count by Age, classified by Education Level.



DISCUSSION



There are trends in the data that show that programming languages are going to be continuously in demand and companies should train their workers to have skills in css,python,javascript,html and type script as they are the future.

The trends are also showing that microsoft sql and postgresql dominate the database market in terms of being used and the future technology prospects.

The data also shows that most respondents are in the age group from 25-34 years old and the least number of people comes from the age group 65 years and older

As people get older they become less engaged in programming showing that it is dominated by very young people.

Most of the respondents have completed their bachelors degree(close to 9000) and about half of them have completed their masters too and only very few respondents only had primary level education .



OVERALL FINDINGS & IMPLICATIONS

Findings

- Java script is one of the most important languages both in the present and future
- Platforms such as amazon web services are increasing in importance over time
- There are trends in the data that show that programming languages are going to be continuously in demand and companies should train their workers to have skills in css, python, javascript, html and type script as they are the future.

Implications

There is great demand for big data skills because the trend of the job market is upward

- There is an increase in importance of advanced programming languages

Companies need to be able to adapt and learn current trends



CONCLUSION



This project demonstrated the need and importance of data analysis in decision making as it allowed for deeper insights on the data to make conclusions that were beyond the surface

The project made use of data analysis techniques such as data extraction using web scrapping and accessing of api keys.

- After extraction came data wrangling skills which include dropping missing data, filling null columns using the appropriate method. Data wrangling also included checking the characteristics of our data to check for outliers in key columns by making use of plots and key python functions.
- After the cleaning of the data to ensure that it can be used for analysis, we engaged in data exploration analysis by making use of different python programmes such as `df.describe`, checking the mean of key columns and the behavior of different variables in the dataset.
- The project was analysis of current and future trends in the stack overflow jo market. It showed the importance of programming languages such as python,sql java,html,css and c#.
- The project strengthened and improves skills in areas like python,sql ,data scrapping data visualization and data exploration
- Inferences in the data will allow for compaines to make informed decisions about the training ,hiring and promotion processes in the business which will help the business be more competitive

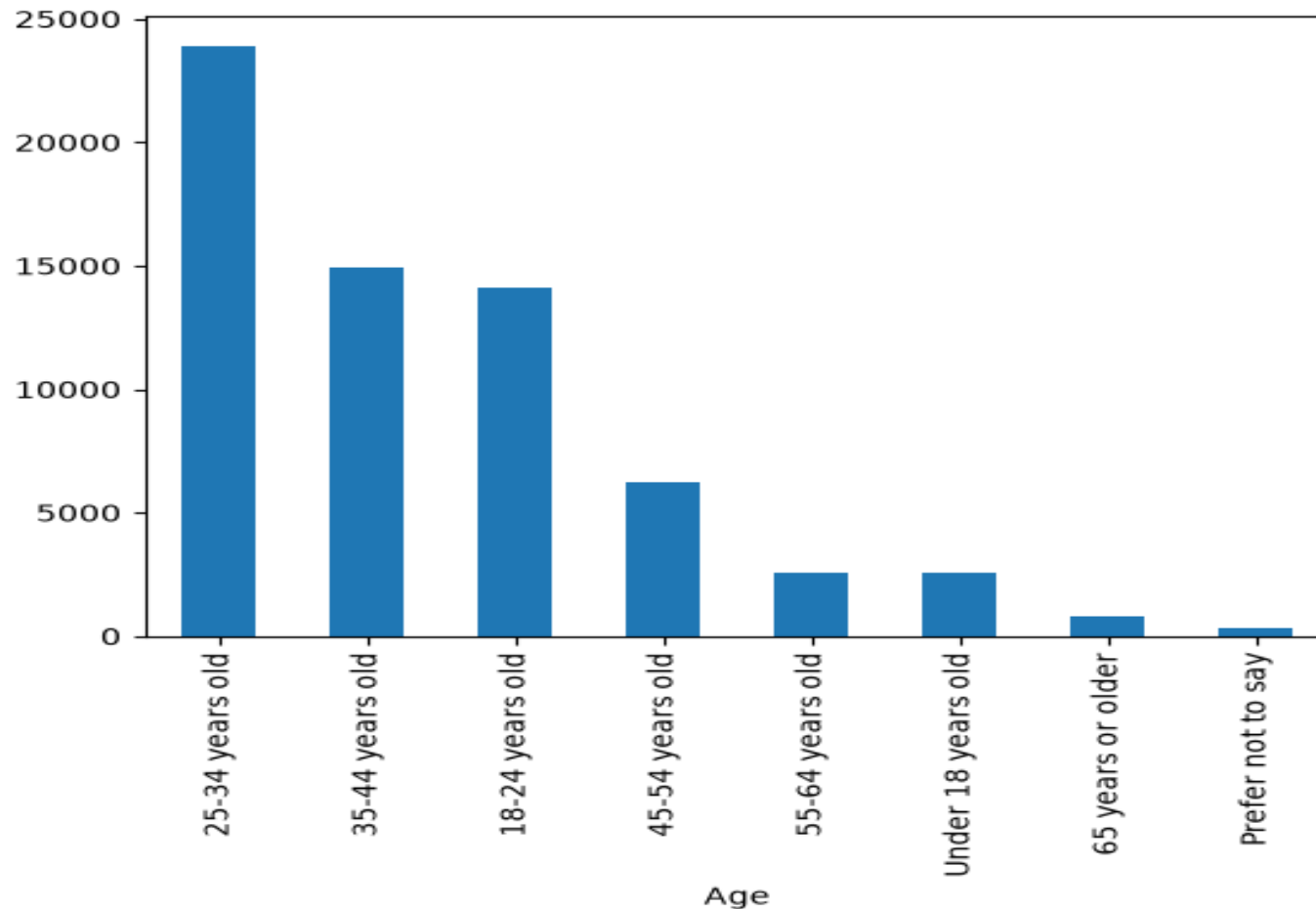


APPENDIX

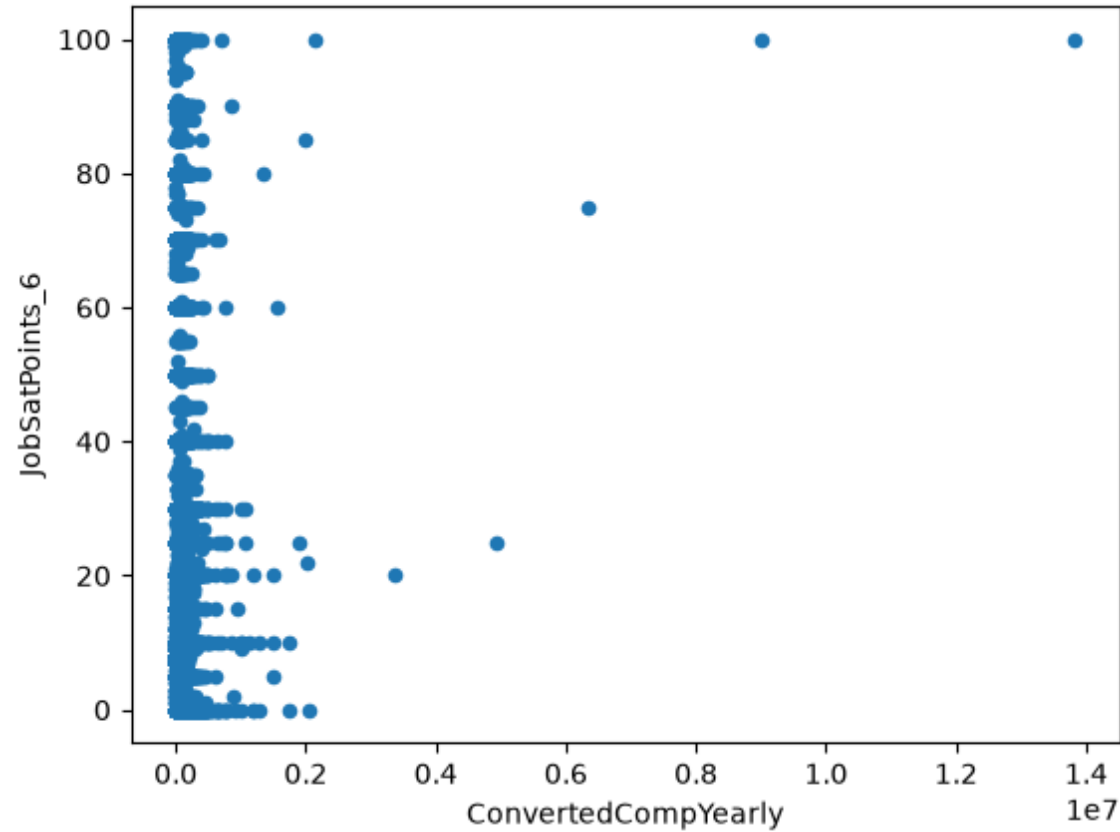


a box plot of Age.

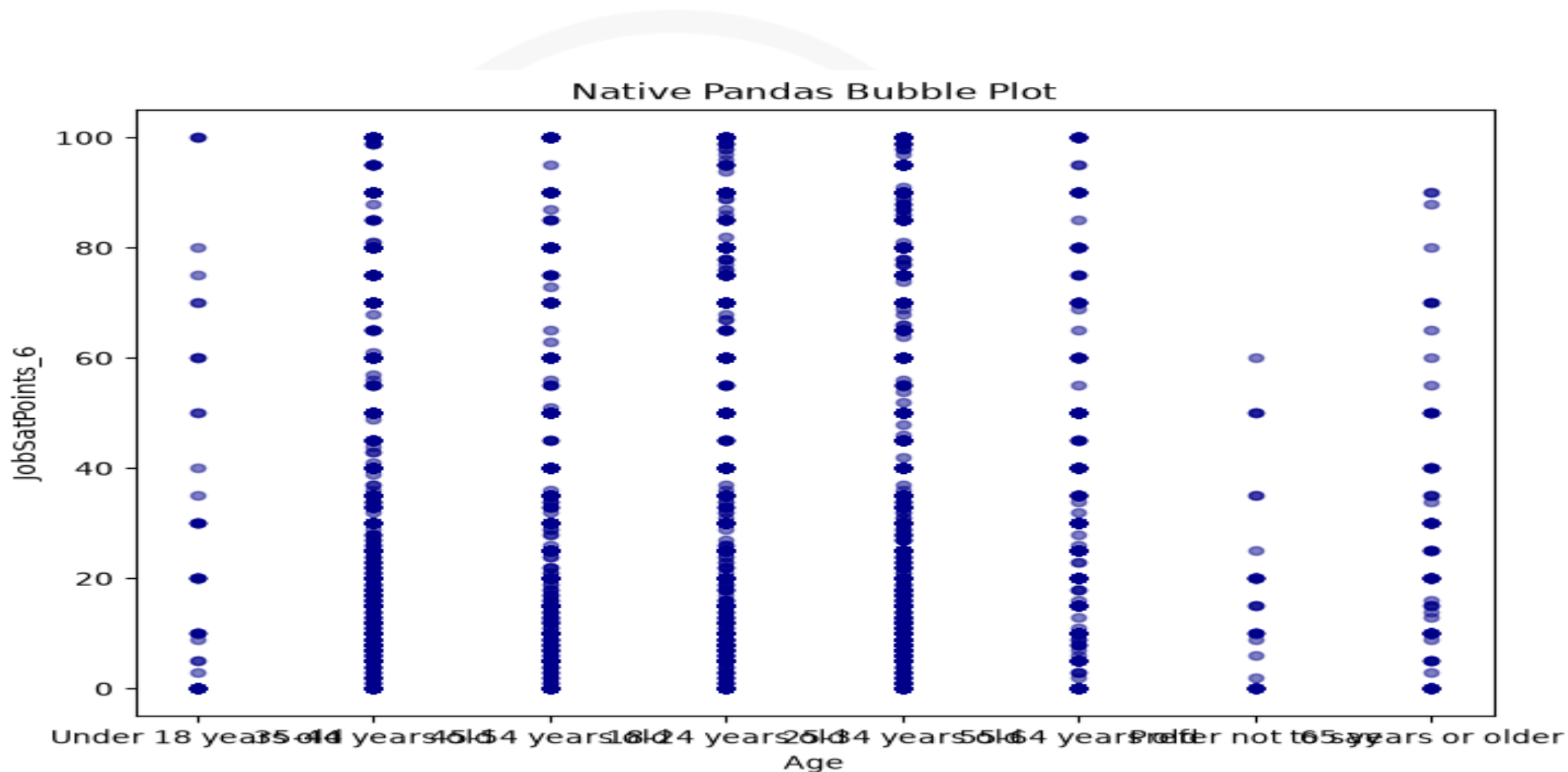
- Extra charts and appendix.



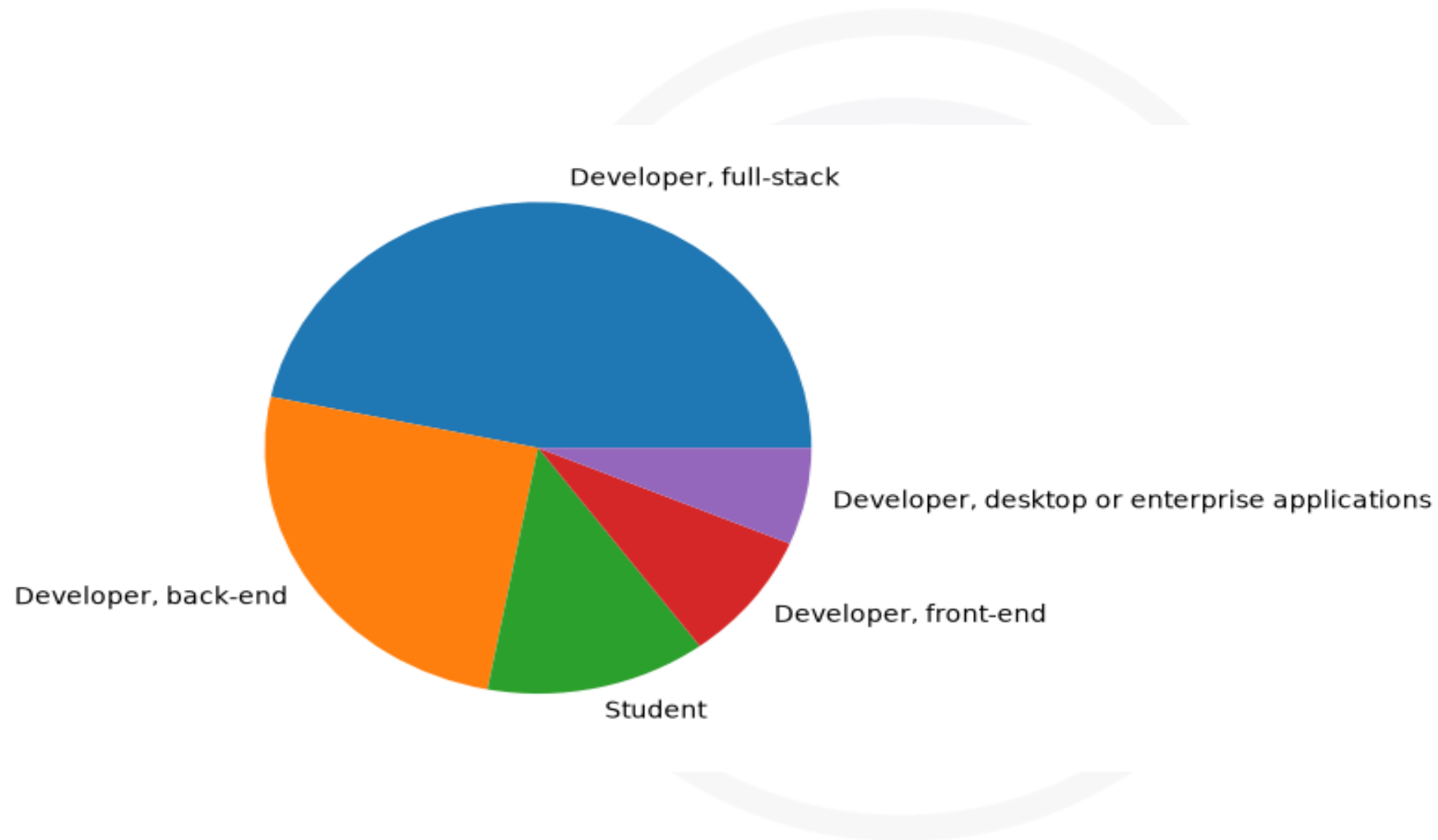
Relationship between job satisfaction and compensation



Bubble plot for age and job satisfaction



Pie chart for developer type



Stacked Chart for Compensation and Job Satisfaction by Age Group

